

Project Title

Adventure Works Sales Analytics Dashboard (Power BI)

1. Project Overview

Objective

The objective of this project is to build an interactive Power BI dashboard that enables business users and management to monitor overall sales performance, analyse sales trends, evaluate product performance, and make data-driven business decisions.

The dashboard is designed to provide executives with a clear view of key business metrics while allowing analysts to drill down into sales and product performance.

2. Business Problem

Adventure Works generates thousands of sales transactions across multiple products and territories.

Business users face challenges in:

- Monitoring overall sales performance
- Identifying profitable products
- Understanding sales trends
- Comparing product performance
- Tracking key business KPIs

A centralized dashboard was required to transform raw sales data into meaningful business insights.

3. Business Objectives

- Monitor overall business performance
 - Track monthly sales trends
 - Identify high-performing products
 - Analyse product profitability
 - Compare sales across territories
 - Support data-driven decision-making
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4. Dataset Information

Dataset Name: Adventure Works Sales Dataset Contains information related to:

- Customers
- Products
- Product Categories
- Product Subcategories
- Sales Orders
- Territories
- Dates

Tables Used:

- Customer
 - Date
 - Product
 - Product Category
 - Product Subcategory
 - Sales Order Header
 - Sales Order Detail
 - Territory
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5. Tools Used

- Microsoft Power BI Desktop
 - Power Query
 - DAX
 - Data Modelling
 - Star Schema
 - Excel Dataset
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6. Skills Demonstrated

- Data Cleaning
- Data Transformation
- Data Modelling
- Relationship Creation

- DAX Measures
 - KPI Development
 - Interactive Dashboard Design
 - Business Analysis
 - Data Visualization
 - Executive Reporting
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7. Data Preparation

Performed the following:

- Imported Adventure Works dataset
 - Validated data types
 - Verified null values
 - Verified duplicate records
 - Created relationships between tables
 - Built a star schema model
 - Created calculated measures using DAX
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8. DAX Measures Created

Include the measures you created (adjust the list if needed):

- Total Revenue
- Gross Profit
- Gross Profit Margin %
- Total Orders
- Total Customers
- Total Quantity Sold
- Average Selling Price
- Total Products
- Active Products Sold
- Total Subcategories
- Revenue Growth
- Revenue YTD
- Previous Month Revenue

- Average Revenue per Customer *(if created)*
 - Average Orders per Customer *(if created)*
 - Average Quantity per Customer *(if created)*
 - Average Gross Profit per Customer *(if created)*
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9. Dashboard Pages

Page 1 – Executive Sales Dashboard

KPIs

- Total Revenue
- Total Orders
- Total Customers
- Total Quantity Sold
- Average Selling Price

Visuals

- Monthly Revenue Trend
 - Revenue by Territory
 - Top 10 Products by Sales
 - Sales by Product Category
 - Territory Slicer
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Page 2 – Sales Intelligence Dashboard

KPIs

- Total Revenue
- Gross Profit
- Gross Profit Margin %
- Average Selling Price

Visuals

- Sales vs Gross Profit by Month
- Monthly Order Trend
- Most Profitable Products
- Sales by Product Subcategory
- Territory Slicer

Page 3 – Product Intelligence Dashboard

KPIs

- Total Products
- Active Products Sold
- Average Product Price
- Total Subcategories

Visuals

- Gross Profit by Product Category
- Average Selling Price by Product Category
- Top 10 Products by Gross Profit
- Top 10 Products by Quantity Sold

10. Key Business Insights

- Revenue trends can be monitored month by month to identify seasonal sales patterns.
- Sales performance varies across territories, helping management identify high-performing regions.
- A small group of products contributes a significant portion of total profit.
- Product categories differ in profitability, supporting better product portfolio decisions.
- High-volume products are not always the highest-profit products, emphasizing the importance of profitability analysis alongside sales volume.
- Average selling price analysis highlights premium product categories and pricing opportunities.

11. Business Value

This dashboard enables management to:

- Monitor sales performance in real time.
 - Identify profitable products and categories.
 - Compare regional sales performance.
 - Support pricing and product decisions.
 - Improve executive reporting through interactive visualizations.
 - Reduce manual reporting effort.
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12. Dashboard Features

- Interactive slicers
 - Dynamic KPI cards
 - Cross-filtering between visuals
 - Responsive dashboard layout
 - Professional corporate design
 - Multi-page navigation
 - Business-focused visualizations
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13. Conclusion

The Adventure Works Sales Analytics Dashboard successfully transforms raw transactional data into meaningful business insights. By combining executive-level KPIs with detailed sales and product analysis, the dashboard supports strategic decision-making, improves reporting efficiency, and provides a comprehensive view of business performance.